SOFTWARE REQUIREMENTS SPECIFICATION (SRS) V1.0

Temar Lije

AI-Powered Hybrid Smart Classroom Platform designed for environments with unreliable internet and electricity.

Problem Statement

Unstable Connectivity

Traditional LMS platforms like Google Classroom fail during frequent internet outages in developing regions.

Power Interruptions

Schools face frequent blackouts, forcing digital learning tools to shut down completely.

High Teacher Workload

Teachers spend hours manually creating quizzes, generating summaries, and tracking attendance.

Limited Out-of-Class Support

Students lack localized tutoring and instant feedback on learning materials outside lecture hours.

Objectives & Core Innovation

Hybrid Edge Hub

Ensures 100% learning continuity by running a local offline classroom server on a teacher's laptop or mini-PC during internet blackouts.

Course-Aware AI

Delivers offline AI tutoring grounded strictly in uploaded teacher materials (PDFs, slides) using lightweight local LLMs.

Auto Cloud Sync

Automatically synchronizes attendance, quiz scores, materials, and messages whenever network connection is restored.

System Users & Roles

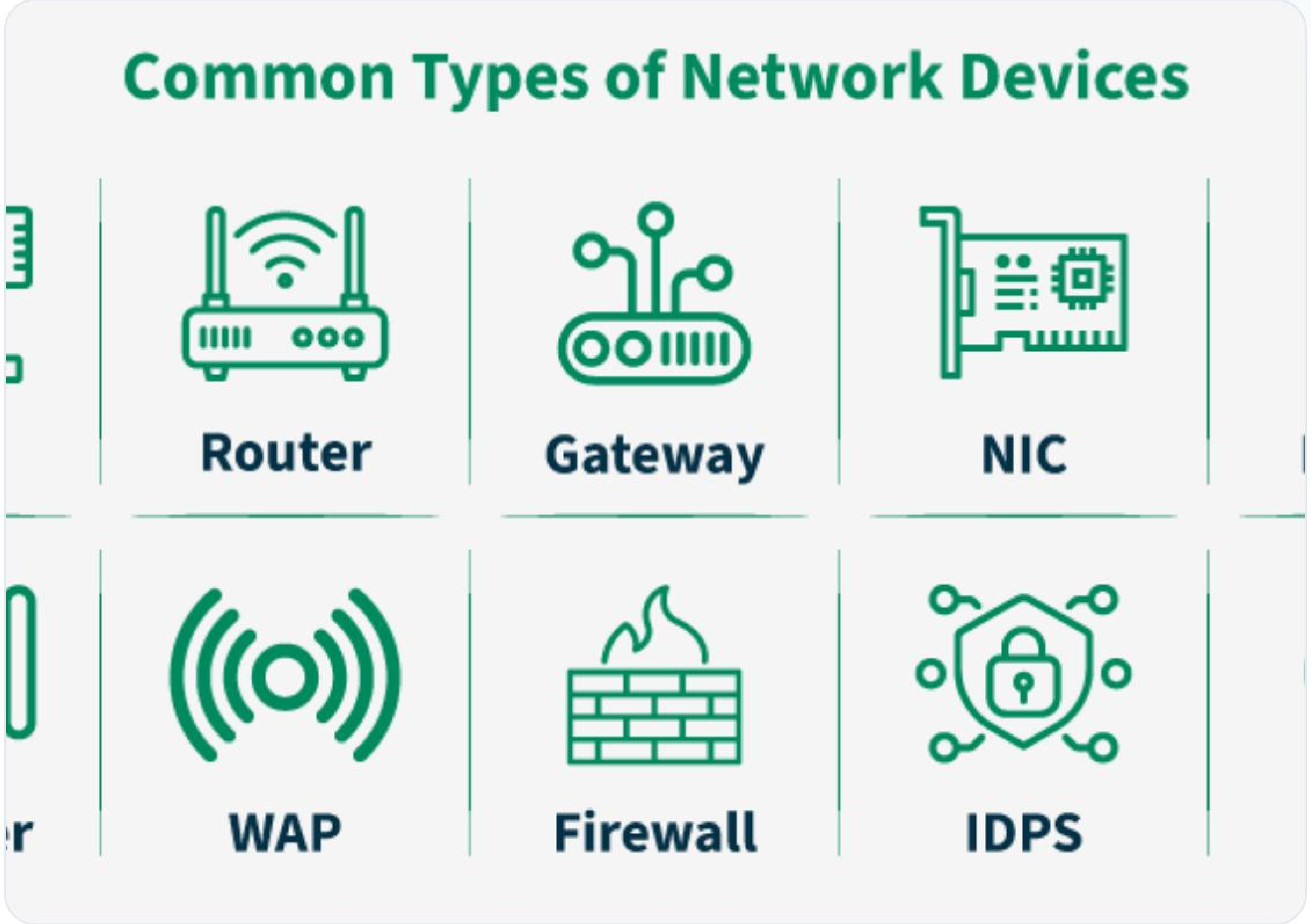
User Role	Primary Responsibilities	Key Platform Capabilities
Teachers	Class management, content delivery, evaluation	Create classes, upload materials, generate AI quizzes & lessons, run live classes.
Students	Coursework execution, active learning	Join via 6-character code, access offline files, submit assignments, consult AI Study Buddy.
Administrators	Institutional oversight & compliance	Manage teacher/student rosters, view classroom analytics, monitor attendance reports.

Hybrid Offline Architecture

Dual Network Execution Engine

Online Mode: Operates on cloud servers (NestJS + PostgreSQL + Cloud AI) for remote access, multi-school sync, and global backup.

Offline Mode: The local Classroom Hub (Teacher laptop or mini-PC) hosts a local server (SQLite + Local AI) providing local Wi-Fi file sharing, quizzes, and live chat without internet.



Key Functional Requirements

Classroom & Content Operations

• Seamless course creation and 6-character invitation code access.
• File repository supporting PDF, PPT, DOCX, and images.
• Integrated video, audio, screen-sharing, and interactive whiteboard.
• Assignment submission with file attachments and grading rubrics.

Dual AI Intelligence Suite

• **Teacher Assistant:** Generates lesson plans, MCQs/True-False quizzes, and post-class summaries.
• **Student Study Buddy:** RAG-based AI tutor answering questions strictly from course documents.
• **At-Risk Detection:** Flags struggling students using attendance and quiz analytics.

Non-Functional Standards

< 3s

Fast System Response

Authentication and material loading times optimized for low-spec hardware.

< 10s

AI Inference Speed

Local and Cloud AI response generation latency strictly capped for smooth interaction.

100%

Offline Reliability

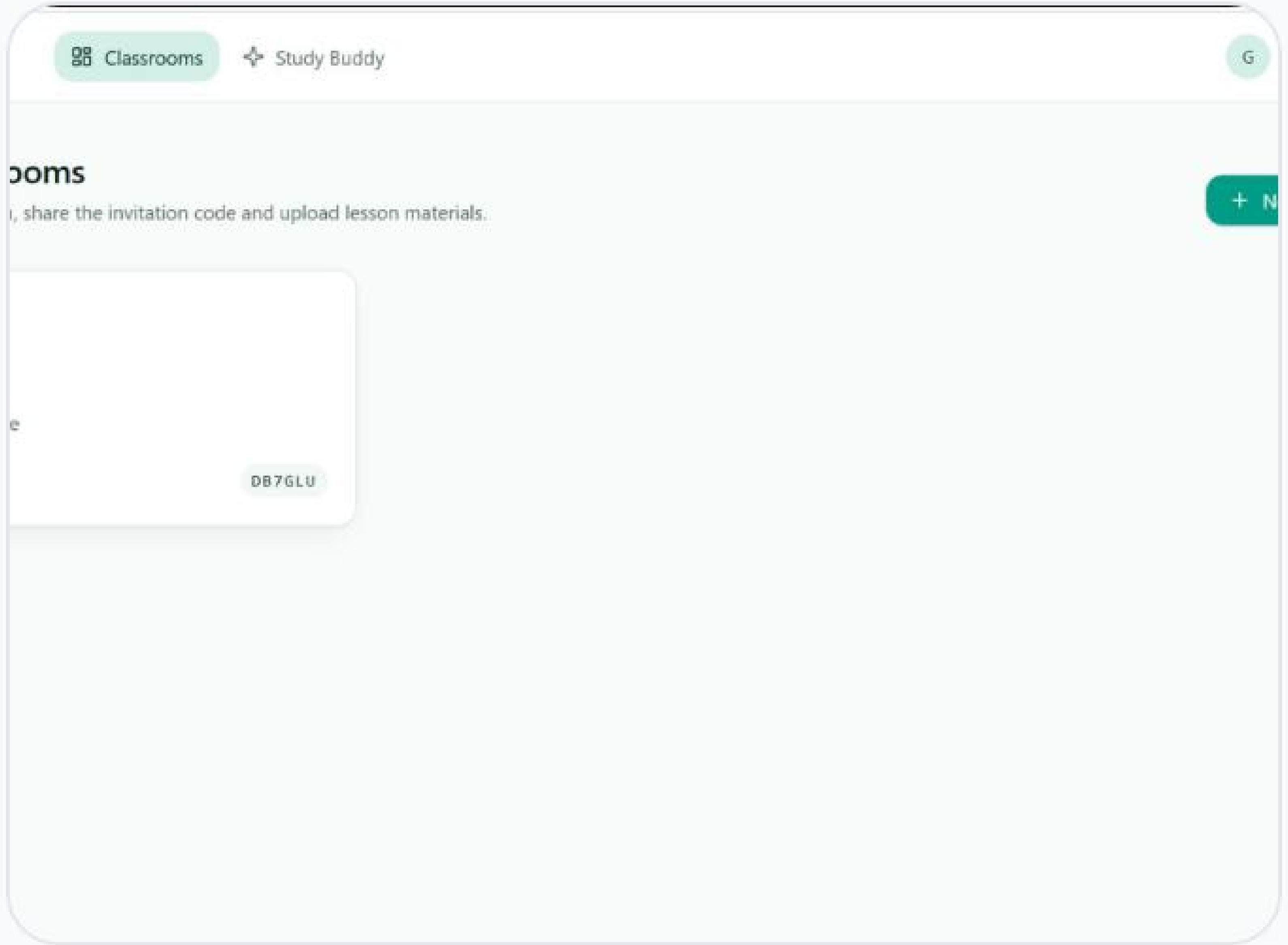
Zero data loss during unexpected power blackouts or network drops.

Technology Stack

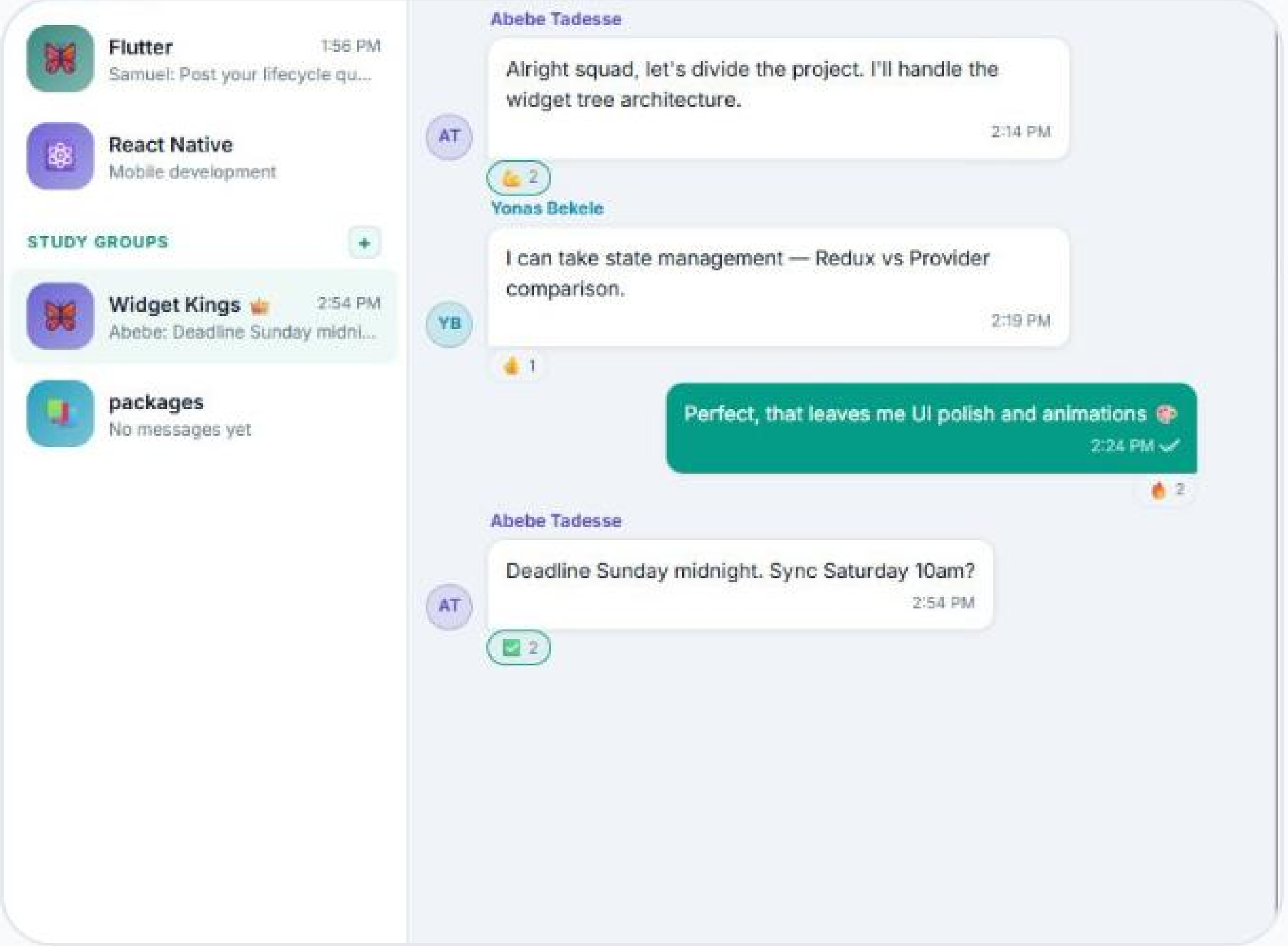
System Layer	Selected Technology	Role & Architectural Purpose
Frontend Client	React + JavaScript (JSX) + Tailwind / PWA	Responsive UI, offline caching via Service Workers & IndexedDB.
Backend Service	NestJS + Node.js	Modular REST API, WebSockets (Socket.io), and sync orchestration.
Database Layer	PostgreSQL (Cloud) / SQLite (Local)	Relational data storage with bidirectional synchronization rules.
AI Engines	Gemini / GPT (Online) & Gemma / Phi (Offline)	Cloud RAG pipeline for online, quantized local LLM for offline RAG.

User Interface Showcase

Landing & Welcome Page



Classroom Dashboard



Real-time Study Groups

Core Competitive Edge

Zero-Internet Hub

Unlike Google Classroom or Moodle, Temar Lije does not require active internet to host live classes, share files, or take quizzes.

Grounded AI Tutoring

Prevents AI hallucinations by constraining Study Buddy responses exclusively to teacher-verified notes and uploaded PDFs.

Intelligent Re-Sync

Queues all local check-ins, quiz submissions, and grades offline, resolving data conflicts automatically upon reconnection.

Limitations & Testing

System Limitations

- Local AI model size restricted (2B-3B parameters) to run on low-spec teacher laptops.
- Initial cloud synchronization requires minimum 2G/3G connectivity.
- Offline video conferencing limited by local Wi-Fi router bandwidth capacity.

Testing & Evaluation

- **Network Interruption Testing:** Simulated sudden network drops during live quizzes.
- **Latency Benchmarking:** Testing offline LLM response times on standard laptops.
- **Conflict Resolution Testing:** Verifying local vs cloud database sync accuracy.

Questions & Answers

Thank you for exploring Temar Lije SRS v1.0

Future Enhancements: Mobile Native App | Speech-to-Text Transcriptions | Multi-School Analytics

Image Sources

https://media.geeksforgeeks.org/wp-content/uploads/20241014175346235579/Common-Types-of-Network-Devices-1.png

Source: www.geeksforgeeks.org